Journal of Structural Biology 203 (2018) 149–161

Contents lists available at ScienceDirect

Journal of Structural Biology

journal homepage: www.elsevier.com/locate/yjsbi

Human skin barrier structure and function analyzed by cryo-EM and molecular dynamics simulation

Magnus Lundborg[a], Ali Narangifard[a,b], Christian L. Wennberg[a,d], Erik Lindahl[c,d], Bertil Daneholt[e], Lars Norlén[e,f,*]

[a] ERCO Pharma AB, Science for Life Laboratory, Solna, Sweden
[b] Department of Medicine, Solna (MedS), Karolinska Institute, Solna, Sweden
[c] Department of Biochemistry and Biophysics, Science for Life Laboratory, Stockholm University, Stockholm, Sweden
[d] Swedish eScience Research Center, Department of Physics, KTH Royal Institute of Technology, Stockholm, Sweden
[e] Department of Cell and Molecular Biology (CMB), Karolinska Institute, Stockholm, Sweden
[f] Dermatology Clinic, Karolinska University Hospital, Stockholm, Sweden

ARTICLE INFO

Keywords:
Stratum corneum
Skin permeation
Cryo-EM
Molecular dynamics simulation
Lipids

ABSTRACT

In the present study we have analyzed the molecular structure and function of the human skin's permeability barrier using molecular dynamics simulation validated against cryo-electron microscopy data from near native skin.

The skin's barrier capacity is located to an intercellular lipid structure embedding the cells of the superficial most layer of skin – the stratum corneum. According to the *splayed bilayer model* (Iwai et al., 2012) the lipid structure is organized as stacked bilayers of ceramides in a splayed chain conformation with cholesterol associated with the ceramide sphingoid moiety and free fatty acids associated with the ceramide fatty acid moiety. However, knowledge about the lipid structure's detailed molecular organization, and the roles of its different lipid constituents, remains circumstantial.

Starting from a molecular dynamics model based on the *splayed bilayer model*, we have, by stepwise structural and compositional modifications, arrived at a thermodynamically stable molecular dynamics model expressing simulated electron microscopy patterns matching original cryo-electron microscopy patterns from skin extremely closely. Strikingly, the closer the individual molecular dynamics models' lipid composition was to that reported in human stratum corneum, the better was the match between the models' simulated electron microscopy patterns and the original cryo-electron microscopy patterns. Moreover, the closest-matching model's calculated water permeability and thermotropic behaviour were found compatible with that of human skin.

The new model may facilitate more advanced physics-based skin permeability predictions of drugs and toxicants. The proposed procedure for molecular dynamics based analysis of cellular cryo-electron microscopy data might be applied to other biomolecular systems.

1. Introduction

1.1. Near-native molecular structure and function of biomolecular complexes in cells and tissues

In vitro experimentation on biomolecular complexes has today reached a high level of sophistication, exemplified by recent progress in cryo-electron microscopy (cryo-EM) single particle analysis (Kühlbrandt, 2014). However, a more complete understanding of biomolecular function may only be achieved by also studying biomolecular complexes directly in their natural environment inside the living cell or tissue. Biological cells, or tissues, are typically crowded multi-component environments lacking long-range order. This makes it difficult to obtain distinct diffraction patterns from inside cells. Nevertheless, access to cellular near-native high-resolution data is today possible through the cryo-EM of vitreous sections technology (Al-Amoudi et al., 2004; Al-Amoudi et al., 2004; Iwai et al., 2012). Here, a major challenge has been the analysis of such cellular cryo-EM data. To this end, electron microscopy (EM) simulation has been employed (Iwai et al., 2012). EM simulation is a way to generate EM images based on

* Corresponding author at: Department of Cell and Molecular Biology (CMB), Karolinska Institute, Stockholm, Sweden; Dermatology Clinic, Karolinska University Hospital, Stockholm, Sweden.
E-mail address: lars.norlen@ki.se (L. Norlén).

https://doi.org/10.1016/j.jsb.2018.04.005
Received 27 March 2018; Received in revised form 19 April 2018; Accepted 22 April 2018
Available online 24 April 2018

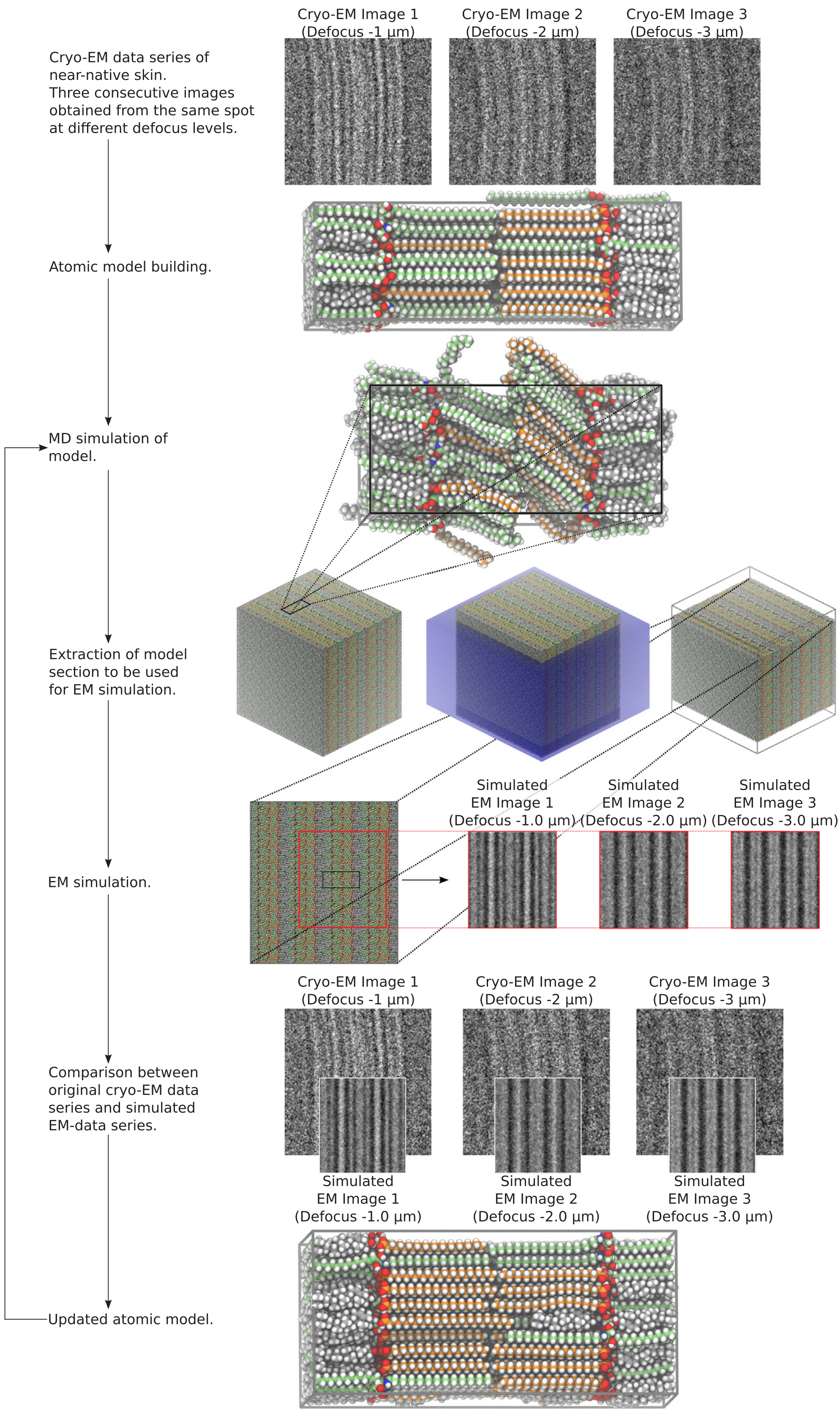

Fig. 1. Experimental procedure. Outline of the general procedure for analyzing the structure and composition of the system by comparing simulated electron microscopy (EM) images, generated from candidate atomic models after MD simulation, with experimental cryo-EM images obtained from near-native human skin, using an iterative approach. The candidate model system was then modified (regarding relative CER, FFA and CHOL concentrations, CHOL distribution, CER EOS content and amount of water in the membrane) until good agreement with experimental data was achieved. In the images depicting molecules all carbon atoms in CER are green, in FFA orange and in CHOL grey. The rest of the atoms are colored according to atom types (oxygen: red, nitrogen: blue, hydrogen: white). Reference original cryo-EM data from skin were obtained at defocus levels of −1, −2, −3 µm and were adapted from Iwai et al. (2012). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the electron scattering properties of an atomistic model and experimental conditions, including features of the electron microscope. However, the full potential of EM simulation-based cryo-EM data analysis may not be reached until standardized procedures for molecular model building have been developed. In this work, we have applied molecular dynamics (MD) simulation for optimizing and standardizing the construction of molecular models used as input structures for electron microscopy simulation. The new procedure has been employed to address one of the more longstanding questions in skin biology—the molecular structure and function of the human skin's permeability barrier.

1.2. Molecular structure and function of the skin's permeability barrier

In the present study we have analyzed the molecular structure and function of the human skin's barrier structure. The skin was developed 360 million years ago to allow the first vertebrates to leave the oceans and adapt to a life on land, by serving as a barrier protecting from dehydration.

The skin's barrier capacity is located to an intercellular lipid structure embedding the cells of the superficial most layer of skin—the stratum corneum. The lipid structure consists of stacked lipid layers composed of ceramides (CER), cholesterol (CHOL) and free fatty acids (FFA) in a roughly molar 1:1:1 ratio.

During the last 40 years much effort has been put into the determination of the molecular organization of the skin's lipid structure, largely driven by the perspective of improving transdermal delivery of drugs. It has long been known that the lipids are organized as stacked layers (Breathnach et al., 1973; Elias and Friend, 1975). Despite the application of a large number of different experimental approaches, including EM (Madison et al., 1987), X-ray diffraction (White et al., 1988, 1991; Garson et al., 1991), electron diffraction (Pilgram et al., 1999), FTIR-spectroscopy (Damien and Boncheva, 2010) and NMR spectroscopy (Björklund et al., 2013), the detailed molecular structure and function of the skin's barrier has remained elusive. Based on these experimental techniques many different structures have been proposed (Swartzendruber et al., 1989; Bouwstra et al., 2001; Hill and Wertz, 2003; McIntosh, 2003; Schröter et al., 2009; Mojumdar et al., 2016). There are also extensive studies on how different lipid compositions, both naturally occurring as well as synthetically modified lipids, influence bilayer features (Eichner et al., 2016; Stahlberg et al., 2016; Stahlberg et al., 2017; Schmitt et al., 2017) and how penetration enhancers interact with the lipids (Pham et al., 2016; Eichner et al., 2017). However, a new technology using cellular cryo-EM *in situ* combined with molecular model building and EM simulation was recently presented making it possible to examine the molecular organization of the skin's barrier in a novel manner (Iwai et al., 2012). It was thus possible to demonstrate that the skin's barrier consists of stacked bilayers of CER in a splayed chain conformation with CHOL associated with the CER sphingoid moiety and FFA associated with the CER fatty acid moiety. The model's lipid arrangement was later supported by infrared spectroscopy studies using deuterated lipids in model membranes (Školová et al., 2014). However, knowledge about the lipid structure's detailed molecular organization, its permeability properties and the individual roles of its different lipid constituents, remains circumstantial.

In the present study we have investigated these properties of the human skin's lipid structure using MD simulation validated against reference cryo-EM data from human skin. We started from the *splayed bilayer model* described above (Iwai et al., 2012) and optimized it further with respect to composition and organization. Finally, the permeability properties and thermotropic behaviour of selected models were evaluated.

1.3. Analysis of cellular cryo-EM data using MD simulation and EM simulation

Atomistic MD simulation combined with EM simulation may be used to analyze cellular high resolution cryo-EM data. Image analysis is then based on an iterative process where the MD model is modified in a stepwise fashion until optimal correspondence is achieved between the original cryo-EM data derived from the biological specimen and the simulated EM data derived from the MD model (Fig. 1).

An advantage, but also a challenge, of using cryo-EM images of cross section slices of stratum corneum, is the high sensitivity of the cryo-EM image interference patterns to the microscope defocus levels.

Using MD simulation to generate input for EM simulation also allows for investigation of the thermodynamic, and other physicochemical, properties of the models. It further ensures a realistic model atomic density irrespectively of model topology. EM simulation errors derived from model artifacts, such as atomic overlapping or void spaces, are thereby avoided.

The procedure for MD/EM simulation-based cryo-EM image analysis is the following: i) Collection of a high resolution cryo-EM image defocus-series. ii) Construction of a candidate atomic model using model building software. iii) Relaxation of the candidate atomic model using MD simulation. iv) Generation of a series of simulated EM images at different microscope defocus levels from the relaxed atomistic MD model using EM simulation. v) Comparison of the original cryo-EM defocus series images and the simulated defocus series images.

The procedure is repeated until optimal correspondence is achieved between simulated and original data (Fig. 1). The simulated EM images, as well as the periodicity of the systems, after MD simulation are shown to be sensitive to the molecular structure and composition, making this a good method for optimizing the systems to achieve high similarity between experimental and simulated EM data.

1.4. Molecular dynamics simulations

Through atomistic MD simulations thermodynamically stable molecular models may be constructed and equilibrated, preferably at very long time scales. The interactions between the atoms of the model are described by biomolecular force fields divided into a bonded (interactions described using bonds, angles and torsion angles) and a nonbonded part (Lennard-Jones and Coulomb interactions). MD simulations may be used to study the molecular properties of a system at a level inaccessible by real-world experiments. However, in order to generate meaningful information the simulated data must be validated against original experimental data. One way of doing this is by comparing simulated EM images derived from atomistic MD models with original cryo-EM images collected from biological cells or tissues.

1.5. Optimization of the skin barrier model

Starting from the lipid barrier model system described by the *splayed bilayer model* (Iwai et al., 2012), the system was optimized in an iterative manner with respect to i) the relative lipid composition (including sphingosine- and phytosphingosine based ceramides, CHOL, FFA, acyl ceramides, cholesterol sulfate, and charged FFA), ii) the distribution of CHOL over the layered structure, iii) the distribution of lipid chain lengths and, iv) the number of water molecules associated with the lipid headgroups. The general procedure for the stepwise optimization of the model's chemical composition and organization is illustrated in Fig. 1. Each model system was subjected to MD equilibration (relaxation) for $\gtrsim$250 ns, followed by a 100 ns production phase, after which it was evaluated against high-resolution cryo-EM image defocus series reference data collected from human skin (Iwai et al., 2012), using EM simulation.

1.6. Electron microscopy simulation

EM images can be simulated from any model—in particular sequences of conformations obtained from MD simulations—by generating an electron scattering potential map from the atomic model and subsequently simulating the interaction between the model's potential map and the electron beam, the optical transformation effect of the electron microscope's lens system and the image formation on the detector. Parameters defining optical properties of the electron microscope, e.g., acceleration voltage, aberration constants and defocus, and the "point spread" function of the detector can be set to mimic the conditions of the real-world cryo-EM experiment (Rullgård et al., 2011).

1.7. Simulation of permeability properties and thermotropic behaviour

The permeability coefficient of a model system to water, or other compounds, can be calculated from the potential of mean force (PMF), i.e. the excess free energy, of the compound along the reaction coordinate crossing the system, and the local diffusion coefficient through the system. These two descriptors can be calculated from nonequilibrium MD simulations by the forward-reverse method, based on Crooks fluctuation theorem (Crooks, 1999; Crooks, 2000; Kosztin et al., 2006).

The thermotropic behaviour of the model system may be estimated by stepwise heating the system, while recording the lipid headgroup area and lamellar periodicity at each temperature step. Disruptions in the linearity of the system dimensions during heating could be indications of minor state transitions in the system. A few systems were selected for studying under heating. They were simulated for 500 ns at each temperature step, in step sizes of 10 K, and step sizes of 5 K in intervals where transitions were expected. The temperature ranged from 303 K to the system melting points (373 K except in one case where it was 383 K).

2. Methods

A more exhaustive description of experimental procedures is available in the Supplementary information.

2.1. Atomic model building

When building the range of candidate model systems based on the *splayed bilayer model* (Iwai et al., 2012) the number of molecules was calculated to make sure that both layers had the same number of molecules of each type. In the first of the two bilayer leaflets each point in the 10 by 10 matrix could be filled by either a CER type molecule or by an FFA or CHOL placed in the CER fatty acid chain region (closer to the membrane center of the built model) and/or a CHOL in the CER sphingoid chain region. FFA and CHOL molecules were aligned so that their polar groups could interact with the CER headgroups. CHOL was located either associated with the CER sphingoid or fatty acid chains. FFA was situated exclusively by the CER fatty acid chains. CHOL and FFA could be placed in the same lattice position if they were placed opposing each other, with the headgroup region in between. The next leaflet was filled by trying to match molecules with long chains in the CER fatty acid region with molecules with short chains in order to create a system with as small gaps as possible between molecules in the two leaflets. The distance between the two bilayer leaflets was set to avoid any collisions.We have used a nomenclature to describe the system composition according to the relative composition in molar %: CER/CHOL/FFA/rel. CHOL on CER sphingoid side/CER EOS/H_2O per lipid. See Fig. S3 for a comparison of the 33/33/33/75/5/0 model system (i.e. a system with 1/3 M fraction each of CER, CHOL and FFA, 75% of the CHOL located by the CER sphingoid chains and 25% of the CHOL located by the CER fatty acid chains, 5% (molar) of the system (i.e. 15% of the total CER fraction) being acyl ceramide EOS (CER EOS) and no water added) before and after MD simulation. CER EOS, if present, was accounted for in the total CER concentration and, with few exceptions, extra FFA of chain length 30 were added to the same amount as the CER EOS, and accounted for in the total FFA concentration, thus affecting the relative concentration of the other FFA chain lengths. In order to keep the number of system components down CER NS, when present, was only included with fatty acid chain length C24 and replaced half of CER NP with fatty acid chain length C24 (the most common chain length); see Tables S1 and S2. CER NP was used as the main CER component in the model systems as it is the most common type of CER in the human stratum corneum (Masukawa et al., 2009). In order to keep the number of CER components down we did not include CER AP, CER AH or CER NH even if they exist in the stratum corneum (Masukawa et al., 2009). The FFA and CER NP fatty acid chain lengths were chosen to represent a realistic distribution (Norlén et al., 1998) (Table S1).

For model systems containing water molecules these were added by a script iteratively running the *gmx solvate* command (Pronk et al., 2013; Abraham et al., 2015) and afterwards removing waters situated far from the headgroup and close to the periodic box edges. The iteration was run with increasing distance from the headgroup, starting at 0.3 nm, until the wanted number of waters per lipid headgroup had been added. The water was not necessarily symmetrically distributed between the two headgroup layers.

2.2. Molecular dynamics simulation

MD simulations were run in GROMACS 5.0 (Pronk et al., 2013; Abraham et al., 2015) using the CHARMM36 lipid force field (Klauda et al., 2010; Venable et al., 2014). The parameters of CER NP were based on CER NS 24:0 and CHARMM36 atom types and bonded parameters and then optimized to reproduce QM torsional data and the crystal structure of the CER (Dahlén and Pascher, 1979). TIP3P (Jorgensen et al., 1983) was used to model water, as it is the standard water model in the CHARMM36 force field. Copernicus (Lundborg and Lindahl, 2014; Pronk et al., 2015; Pouya et al., 2017), was used to run free energy calculations of water (solvation and binding free energy calculations to get the free energy of inserting the studied molecule in the intercellular lipid structure) as well as the forward-reverse pulling simulations through the lipid systems. Copernicus was also used to distribute work to multiple computers over a network.

All models of the intercellular lipid structures were subjected to energy minimization, equilibration and production runs. Equilibrations were run in five different stages for a total equilibration time of approximately 270 ns, 250 ns of which were without restraints. The production runs lasted 100 ns. The temperature was set to 303.15 K during equilibration and production simulations. During the production phase there were no large drifts in potential energy or system dimensions for any of the systems leading to the conclusion that the systems were largely relaxed. For more details regarding the MD simulations see S1.2 in the Supplementary information.

All images representing molecules were prepared using Tachyon (Stone, 1998) in VMD (Humphrey et al., 1996).

2.3. Electron microscopy simulation

Simulated EM images were generated from the lipid model systems, based on the coordinates of the last frame from MD simulations, using an EM simulation program, called TEM-simulator, by Rullgård et al. (2011).

A virtual specimen was attained by repeating the MD periodic box until it was large enough to be comparable with a real specimen in terms of its thickness (~30 nm) and the number of visible repeating bands. The parameters of the EM simulation were set to match the nominal values recorded during the actual microscopy, i.e. the defocus value, dose (3300 e/Å^2), magnification (85,000) and the detector's pixel size (16 µm). System compositions were assessed by visual comparison of simulated EM images and the high-resolution reference cryo-electron micrographs (Iwai et al., 2012) at three different defocus levels: −1.0, −2.0, −3.0 µm.

EM simulations were also run at other defocus levels, ranging from 0.0 µm to 5.0 µm, with steps of 0.1 µm, as the nominal defocus levels of the experimental micrographs might differ from the actual defocus levels by up to about 1 µm in cryo-EM of vitreous sections. When comparing the simulated EM images with the original defocus series the image at defocus −1 µm was scaled down by 15% and the image at −2 µm by 11% to make the periodicity the same in the defocus series. This scaling was performed in all figures displaying the original reference cryo-EM images from Iwai et al. (2012).

2.3.1. Water permeability

Water permeability was calculated from the PMFs and the local diffusion coefficients throughout the systems, obtained using a nonequilibrium forward-reverse pulling method (Crooks, 1999; Crooks, 2000; Kosztin et al., 2006; Chen, 2008) starting from the final frame of the production MD simulations. The detailed procedure for calculating the water permeability is described in the Supplementary information (S1.2.2).

2.3.2. Heating

Five representative systems were heated from 303 K to 373 K in steps of 10 K. Each step except the first started from the previous temperature in the interval. In addition to these temperatures 338, 348, 358, 368 K were also run, starting from 333, 343, 353, 363 K, respectively. The simulations were run for 500 ns each. During the heating series the same simulation parameters were used as before except that the two bilayer leaflets in the periodic system were assigned to separate groups for temperature coupling and center of mass motion removal. Comparing the simulation at 303 K with previous simulations verified that this did not make any difference for the results. The first 25 ns of the simulations were discarded as equilibration at each temperature.

3. Results

3.1. Optimization of the model system

After MD simulation, the *splayed bilayer model* (Iwai et al., 2012) had a periodicity of 8.5 nm, which was shorter than the reported 11 nm (Iwai et al., 2012; Moghadam et al., 2013). The system (Iwai et al., 2012), was optimized by first introducing a more realistic ceramide fatty acid, and free fatty acid, chain length distribution varying between C20–C30 and peaking at C24–C26, based on that of native human skin (Norlén et al., 1998) (Table S1). Replacing the monodistributed C24 lipid chains with the broad skin lipid chain length distribution resulted in an increased lamellar periodicity of the system from 8.5 nm of the original system (Fig. 2) to 9.2 nm (Table S3; Fig. S6a). Subsequently, the model was improved in an iterative manner with respect to i) CHOL distribution over the bilayer structure (the fraction located in the CER sphingoid region with the rest located in the CER fatty acid region), ii) CER EOS content, iii) water content, iv) CER, CHOL and FFA content, v) CER headgroup composition, vi) CER sphingoid chain length, vii) cholesterol sulfate content and viii) FFA charge.

Using cryo-EM, orderly arranged molecular assemblies, like the stratum corneum lipid structure, give rise to interference patterns when examined in defocus at high magnification. By recording a series of images at different microscope defocus levels at the same position of the skin sample, a series of interference patterns can be obtained, and thus increase the discriminative power of the subsequent EM simulation based image analysis. The interference patterns of cryo-electron micrographs are most sensitive to the local amount and distribution of heavy atoms such as nitrogens and oxygens. In the stratum corneum lipid structure, these heavy atoms appear in the lipid headgroups, in CHOL, in the ester bonds of the acyl ceramides, and in water. Accordingly, we started the optimization process by varying the CHOL distribution over the lipid bilayer structure (Figs. 3a; S6–S7; Tables S3–S4), as well as the CER EOS content (Figs. 3b; S8–S9; Tables S5–S6) and the number of water molecules associated with the lipid headgroups (Figs. 3c; S10–S11; Tables S7–S8). Subsequently, we varied the system's total CER content (Figs. 3d; S12–S13; Tables S9–S10), total CHOL content (Figs. 3e; S14–S15; Tables S11–S12) and total FFA content (Figs. 3f; S16–S17; Tables S13–S14). Finally, we varied the system's CER headgroup composition (sphingosine- and phytosphingosine-based, respectively), CER sphingoid chain length distribution (C18 and C20, respectively), cholesterol sulfate content and FFA charge (degree of deprotonation) (Figs. S18–S21; Table S15–S18).

Cholesterol distribution. By distributing 100% to 60% CHOL on the CER sphingoid side (0% to 40% on the CER fatty acid side) (Fig. S6) the periodicity of the model system changed from 9.2nm (100%) to 10.0 nm (60%), peaking at 10.5 nm for the models with 75% or 70% of the CHOL situated on the CER sphingoid side (Table S3). Simulated EM images from models with 70 to 80% CHOL situated on the CER sphingoid side agreed best with original cryo-EM data (Figs. 3a; S7; Table S4).

Acyl ceramide content. A molar content of CER EOS in the range of 0%–15% of the total lipid content (Fig. S8) resulted in an increased periodicity of the model system from 10.0 nm (0%) to 11.2 nm (15%) (Table S5). Optimal correspondence between simulated and original micrographs over the whole defocus range was obtained for the model with a CER EOS content of 5%. Models with 3% or 8% CER EOS content corresponded less well with the original cryo-EM data (most clearly visible at the defocus of −2 µm) (Figs. 3b; S9; Table S6). An equal number of FFA to that of CER EOS, were prolonged to C30 chain length (Table S2). Increasing the C30 FFA fraction did not affect the periodicity of the system, but improved the similarity between the simulated and experimental electron micrographs (Figs. S20–S21; Tables S17–S18).

Water content. Increasing the number of water molecules associated with the lipid headgroups from 0 to 2 (Fig. S10) resulted in a slightly increased periodicity of the model system from 10.5 nm (0 water molecules/lipid headgroup) to 10.8 nm (2 water molecules/lipid headgroup) (Table S7). In the simulated electron micrographs, raising the number of water molecules associated with the lipid headgroups from 0 to 2 had a strong effect on the band patterns (sharpness and width) and band intensities, but not on the band positions (Figs. 3c; S11). Visual and numerical comparison, of the simulated EM images with the experimental data, ranked the model with 0.3 water molecules per lipid headgroup highest (Figs. S11; Table S8).

Total ceramide content. Varying the total molar CER content, of the model systems, from 10% to 60% (Fig. S12) resulted in a periodicity change of the model system from 10.1 nm (10% CER) to 9.9 nm (60% CER), peaking at a periodicity of 10.6 nm for the model systems containing 33% and 35% CER, respectively (Table S9). In the simulated electron micrographs, increasing the CER content from 10% to 60% had a strong effect on the band patterns (sharpness and width) and band intensities (Figs. 3d; S13). The best agreement between simulated and original micrographs was obtained for the models with 33% and 35% CER content (Fig. S13; Table S10).

Total cholesterol content. In the range from 10% to 50% total molar CHOL content (Fig. S14) the periodicity of the model system changed from 10.1 nm (10% CHOL) to 8.8 nm (50% CHOL), peaking at a periodicity of 10.8 nm for the model system containing 35% CHOL (Table S11). Optimal similarity between simulated and original EM images was obtained for models with 33% and 35% CHOL content (Figs. 3e; S15; Table S12).

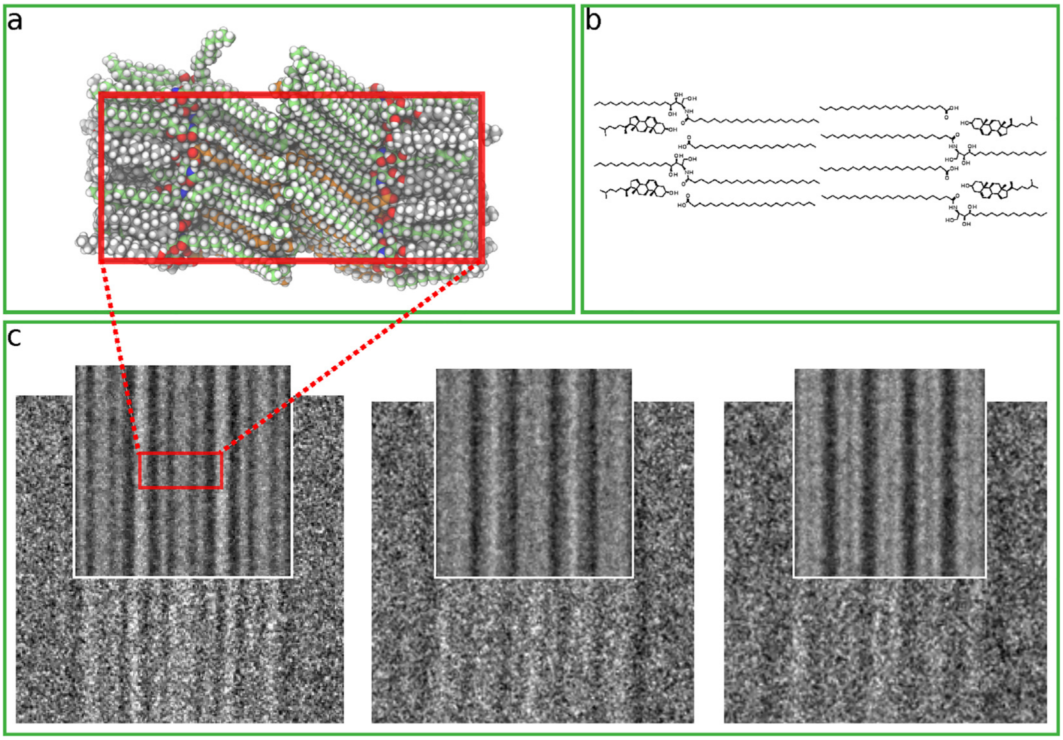

Fig. 2. The *splayed bilayer model* (Iwai et al., 2012). a) shows the system after the MD simulation production phase. In b) can be seen a molecular drawing of the proposed arrangement of the system. In c) a comparison between experimental (beneath) and simulated (above) EM images at defocus levels of −1, −2, −3 μm is shown. In the images depicting molecules all carbon atoms in CER are green, in FFA orange and in CHOL grey. The rest of the atoms are colored according to atom types (oxygen: red, nitrogen: blue, hydrogen: white). The red box in the molecular system represents the periodic box, or unit cell, used in the MD simulations. Atoms outside the periodic box appear on the opposite side. Reference original cryo-EM data from skin were obtained at a defocus levels of −1, −2, −3 μm and were adapted from Iwai et al. (2012). The reference images at defocus levels −1, −2 μm have been scaled down by 15 and 11%, respectively, to keep the same periodicity at all defocus levels. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Total free fatty acid content. Varying the total molar FFA content, of the model systems, from 20% to 50% (Fig. S16) resulted in a periodicity change of the model system from 9.3 nm (20% FFA) to 9.7 nm (50% FFA), peaking at a periodicity of 10.6 nm for the model systems containing 33% to 40% FFA, respectively (Table S13). Visual and numerical comparison of the EM images ranked the model system with 33% FFA content highest overall (Figs. 3f; S17; Table S14).

Other modifications. Varying the CER headgroup composition, CER sphingoid chain length, cholesterol sulfate and FFA charge did not have a significant effect on the periodicity of the model systems (Tables S15 and S17). As judged by visual inspection all the above compositional variations reduced the correspondence between simulated and original data (Figs. S19 and S21). The difference was, however, subtle for the variations in CER sphingoid chain length (most clearly visible at the defocus of −2 μm, see Fig. S21). Numerical comparison also showed a reduced correspondence for all the compositional variations and agreed well with visual inspection (Tables S16 and S18). In the model system containing charged FFAs, the amount of 10% charged FFAs is unrealistically high. 10% charged FFAs was chosen in order to see if the FFA protonation state could affect the simulated electron micrographs and the calculated permeability coefficients of water at all. The reported pK_a of FFAs in bilayers of stratum corneum lipid mixtures and other lipid bilayers range from approximately 6 (Lieckfeldt et al., 1995; Gómez-Fernández and Villalaín, 1998) to 10 (Schullery et al., 1981). It is expected that the pK_a of the FFAs in the stratum corneum lipid structure is at least equally high, as it contains very little water (Iwai et al., 2012).

3.2. Water permeability

The intercellular lipid structure in the stratum corneum acts as a barrier to prevent water loss, but is not completely impermeable to water. A desired property of an idealized computer model is a water permeability that is lower than what is experimentally observed in diffusion cells, in which the skin membrane is hydrated and possibly has local defects. The permeabilities to water of representative model systems, including the 33/33/33/75/5/0.3 system, are presented in Table 2. Compositional variations consisted of their i) lipid chain length distribution, ii) cholesterol distribution over the bilayer structure, iii) CER EOS content, iv) water content, v) CER headgroup composition, vi) CER sphingoid chain length, vii) cholesterol sulfate content, viii) charged FFA content, ix) added FFA 30 content as well lipid bilayer systems with CER in hairpin conformation, with and without water between the studied bilayers.

Calculated water permeability coefficients are presented in Table 2. The presented uncertainties of the calculated log K_P (the logarithm of the skin permeability coefficient) values are the statistical standard errors. They do not take into account that the simulations have not covered all possible routes through the system. It is therefore possible that the true standard errors are higher.

What could be seen from the water permeability simulations is that introducing a chain length distribution varying between C20 and C30, peaking at C24–C26, reduced the water permeability of the model system, while increasing the relative amount of CHOL associated with the CER sphingoid side did not change the water permeability significantly, and that adding CER EOS increased the water permeability. The effect of adding water to the system was nonlinear. Low amounts of water in the system (0.3–0.7 water molecules per lipid), by the lipid headgroups, reduced the water permeability. Highest permeability was

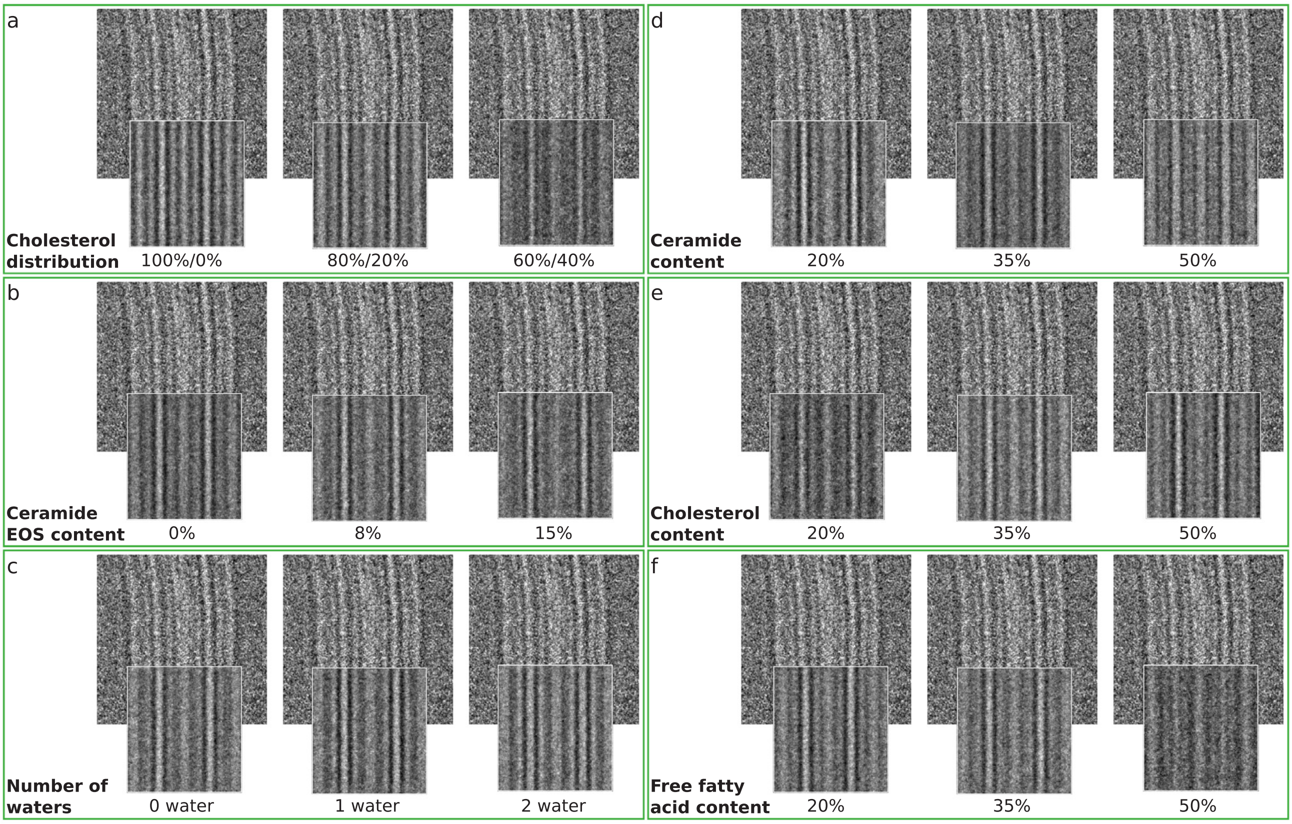

Fig. 3. Effect of CHOL distribution, CER EOS concentration, amount of water per lipid and the CER, CHOL and FFA concentration on the simulated EM data. A comparison between a cryo-EM image obtained from near-native skin at defocus -1 µm (larger upper set of images, adapted from Iwai et al. (2012)) and simulated EM images (defocus -1.0 µm) from MD models with various compositions. a) CHOL distribution over the lipid bilayer system (% CHOL on CER sphingoid side/% CHOL on CER fatty acid side). b) CER EOS content of total lipid content. c) Number of water molecules per lipid headgroup. d) CER content of total lipid content. e) CHOL content of total lipid content. f) FFA content of total lipid content. For the complete sets of data for a)–f) see Figs. S7, S9, S11, S13, S15 and S17.

recorded with 1.3 water molecules per lipid or very high amounts of water (3 water molecules per lipid). Nonlinear effects of water content in skin and phospholipid membranes, on permeability and transepidermal water loss (TEWL) have been reported before (Grice et al., 1972; Blank et al., 1984; Sparr and Wennerström, 2001), but not the herein observed multiple local minima and maxima in the permeability coefficient depending on water content. No difference could be seen between the two model systems with only NS or only NP CER, respectively, and also no distinction between these models and the model with a mixture of NS and NP CER. With 10% of the CHOL replaced by cholesterol sulfate the permeability increased significantly.

The '33-hairpin/33/33 bilayer with water' is a bilayer system with water on each side and CER C24 and FFA C24. This system is comparable to what Das et al. and Gupta et al. used in their permeability calculations, but with only CER NP instead of CER NS (Das et al., 2009; Gupta et al., 2016). At 310 K Gupta et al. calculated a log K_P of $-0.6(1)\,\mathrm{cm\,h^{-1}}$, which at 303 K would be approximately $-0.9(1)\,\mathrm{cm\,h^{-1}}$, using the same correction as Abraham and Martins (Abraham and Martins, 2004). In the simulations by Das et al. the water $logK_{P_{350K}}$, in cm/h, was roughly 0.5 (recalculated from reported ΔG and D(z) values) (Das et al., 2009), which would correspond to $logK_{P_{303K}}$ of about -1.4 with the same temperature corrections as above (Abraham and Martins, 2004) (provided that the temperature dependence is linear over the large temperature range). Our calculations give a similar permeability, considering different types of ceramides, force fields and methods, with a log K_P of $-1.4(1)\,\mathrm{cm\,h^{-1}}$.

The proposed optimized system (33/33/33/75/5/0.3) had a simulated water permeability of log K_P $-4.5\,\mathrm{cm\,h^{-1}}$. This is lower than that of human skin in *in vitro/ex vivo* diffusion cell experiments (approximate log K_P of $-2.9\,\mathrm{cm\,h^{-1}}$ at 303 K (Scheuplein, 1965; Scheuplein and Blank, 1973; Abraham and Martins, 2004)).

The PMFs and diffusion profiles, of water through the proposed model system, referred to as 33/33/33/75/5/0.3, are presented in Fig. S22.

3.3. Thermotropic behaviour

The five tested model systems, that were all based on CER and FFA with a broad fatty acid chain length distribution, varying between C20–C30 peaking at C24–C26, and with a 1:1:1 M ratio between CER, FFA and CHOL, but with varying CHOL distribution, CER EOS content and water content, behaved similarly when heating from 303 K to 373 K. Supp. Fig. S23 shows how the area per CER (system area divided by the number of CER, i.e., implicitly including area contribution from CHOL and FFA) and periodicity of the systems changed upon heating. The thermotropic T1 transition is supposedly due to phase changes in the lipid bilayer (Babita et al., 2006). It is difficult to determine if this transition occurs in the MD simulations of the different systems, but the area and periodicity plots indicate a slight change in the 33/33/33/100/5/0.3 and the 33/33/33/75/5/0.3 model systems. All systems showed changes in the area and periodicity plots in the regions of T2 and T3, both of which are assumed to arise from lipids, possibly from melting of different lipid components (Babita et al., 2006). All five simulated model systems melted in the T4 temperature region, which has previously been assigned to denaturation of a protein component (Babita et al., 2006).

3.4. Comparison with conventional RuO_4-staining electron microscopy data

The electron density distribution of the optimized skin lipid model resembled the ruthenium tetroxide staining pattern of the lipid structure in skin (Madison et al., 1987) (Fig. 5). The model's lipid headgroups give rise to the thick dark bands, whereas the ester groups of the CER EOS give rise to the thinner and more diffuse dark bands.

4. Discussion

Starting from a molecular dynamics model based on the splayed bilayer model (Iwai et al., 2012), we have, by stepwise structural and compositional modifications, arrived at a thermodynamically stable molecular dynamics model for the stratum corneum lipid structure expressing simulated electron microscopy patterns matching original cryo-electron microscopy patterns from skin extremely closely.

This experimental approach, based on comparing cellular cryo-EM images collected at several different microscope defocus levels with simulated EM images derived from atomic models equilibrated, or relaxed, by MD simulation, is highly discriminative, and may allow for subnanometer resolution structural and functional analysis of biomolecular assemblies directly inside near-native cryo-immobilized cells and tissues. For example, we have shown that the stratum corneum's lipid structure likely contains less than one water molecule per lipid headgroup (Figs. 3c; S10–S11; Tables S7–S8) and that water may have an effect on the lipid structure's calculated permeability (Table 2). Increasing the model's relative amount of CER EOS does increase the lipid structure's calculated permeability even more than does increasing above 1 its number of water molecules per lipid headgroup, while changing the distribution of CHOL over the model's bilayer structure does not seem to have a significant effect on permeability (Table 2). Furthermore, introducing a broad chain length distribution in the model's CER fatty acid and FFA fractions, similar to that of stratum corneum lipid extracts, does reduce the water permeability (Table 2).

4.1. Model periodicity

By running MD simulations we have shown that the *splayed bilayer model* (Iwai et al., 2012), with only one type of CER component (C24-ceramide NP) and all CHOL associated with the CER sphingoid moiety has a lamellar periodicity of approximately 8.5 nm. Thereby it is not extended enough to reproduce the approximate 11nm periodicity reported in recent cryo-EM (Iwai et al., 2012) and small-angle X-ray scattering (Moghadam et al., 2013) experiments. However, apart from the lamellar periodicity, the general features of the simulated EM images, also after MD simulation, are in fair agreement with experimental data from skin (Fig. 2; Table 1). A more complex, and realistic (Norlén et al., 1998), composition of CER fatty acid- and FFA chain lengths (Tables S1–S2), together with the addition of CER EOS gives a periodicity closer to the experimental 11 nm values (Fig. 4; Table 1). A lamellar periodicity of 13 nm, instead of 11 nm, has previously been reported by some groups using small-angle X-ray diffraction (SAXD) on stratum corneum isolated from mouse, pig and human skin (White et al., 1988; Bouwstra et al., 1991; Hou et al., 1991; Bouwstra et al., 1995; Ohta et al., 2003; Hatta et al., 2006; Nakazawa et al., 2012) and on skin lipid model mixtures (Bouwstra et al., 1998; Bouwstra et al., 2001; Bouwstra et al., 2002; de Sousa Neto et al., 2011; Mojumdar et al., 2013; Mojumdar et al., 2015; Mojumdar et al., 2016), but not by other groups (Garson et al., 1991; Moghadam et al., 2013). We do not yet have an explanation for the reported 13 nm periodicity. However, we found that model systems consisting of only CER EOS had a periodicity of 13.0 nm after MD simulation. This was irrespectively if they were organized as a bilayer with the CER EOS fatty acid moieties oriented towards each other and the CER EOS sphingoid moieties oriented towards each other, or as a stacked monolayer with the CER EOS fatty acid moieties oriented towards the CER EOS sphingoid moieties (Fig. S24).

Table 1
Periodicity and rank of model systems after MD simulation. The periodicity was measured during the 100 ns production phase, after relaxation of the systems. The rank by automatic comparison was calculated using the numerical difference (after scaling the intensity, but not the width, and moving) of the intensity curves. The rank by visual inspection is based on a comparison of the images by eye. The rank is shared if the difference is low (see the Supplementary information for details). Shared ranks leave a gap afterwards. For the optimized model systems the relative composition is specified in molar % (../../../../../..): (total CER content/total CHOL content/total FFA content/relative amount of CHOL on CER sphingoid side/total CER EOS content/number of H_2O per lipid headgroup).

ID	Periodicity (nm)	Rank by automatic comparison	Rank by visual inspection
Splayed bilayer model (Iwai et al., 2012)	8.5	7	8
33/33/33/100/5/0.3	9.2	7	5
33/33/33/75/0/0.3	10.0	4	2
33/33/33/75/15/0.3	11.2	5	6
33/33/33/75/5/0	10.5	1	2
33/33/33/75/5/1	10.8	3	2
33/33/33/75/5/2	10.8	5	6
33/33/33/75/5/0.3	10.6	1	1

4.2. Model lipid composition

The relative molar lipid composition reported in literature is approximately 1:1:1 of CER:CHOL:FFA (Weerheim and Ponec, 2001). Our results support that the molar fractions of CER, CHOL and FFA likely are about 33% each, or around 33–35%, and also show that the total CER EOS content in the lipid structure likely is about 5 M%, with a plausible range of 3–8 M%, which is slightly higher than what has been reported (Masukawa et al., 2009). Speculatively, this may be explained by that the large and bulky long-chain acyl ceramides may be more difficult to extract completely from skin samples than the other lipid species and thus slightly underestimated quantitatively.

CHOL is shown to be inhomogeneously distributed over the stratum corneum lipid structure with about 25%, or in the range 20–30%, associated with the CER fatty acid chains and the rest with the CER sphingoid chains.

According to our analysis, the stratum corneum lipid structure's water content is low, confirming what has previously been reported (Bouwstra et al., 2003; Iwai et al., 2012), being approximately 1/3 water molecules per lipid headgroup. This corresponds to a water concentration of 0.65M in the lipid structure. This value is lower than reported water concentrations in the stratum corneum as a whole, estimated to be 4 M to 50 M, depending on the location in the stratum corneum (Blank et al., 1984; Potts et al., 1985). This is in agreement with the notion that water is not homogeneously distributed in the stratum corneum, with the bulk water situated intracellularly in the corneocytes, and that only a small fraction, if any, is situated by the lipid headgroups in the intercellular space (Bouwstra et al., 2003; Iwai et al., 2012). Based on our results it is possible that there is more than 1/3 water molecules per lipid headgroup, but definitely not more than one water molecule per lipid.

Interestingly, increasing the model's relative amount of C30 FFA with respect to remaining FFA, in order to reach a molar 1:1 ratio between C30 FFA and CER EOS, increased the correspondence between simulated and original cryo-electron micrographs (Fig. S21). Precise quantitative analysis of the C30–36 FFA and ceramide fractions extracted from human stratum corneum would be called for to verify experimentally the here tentatively suggested presence of such a 1:1

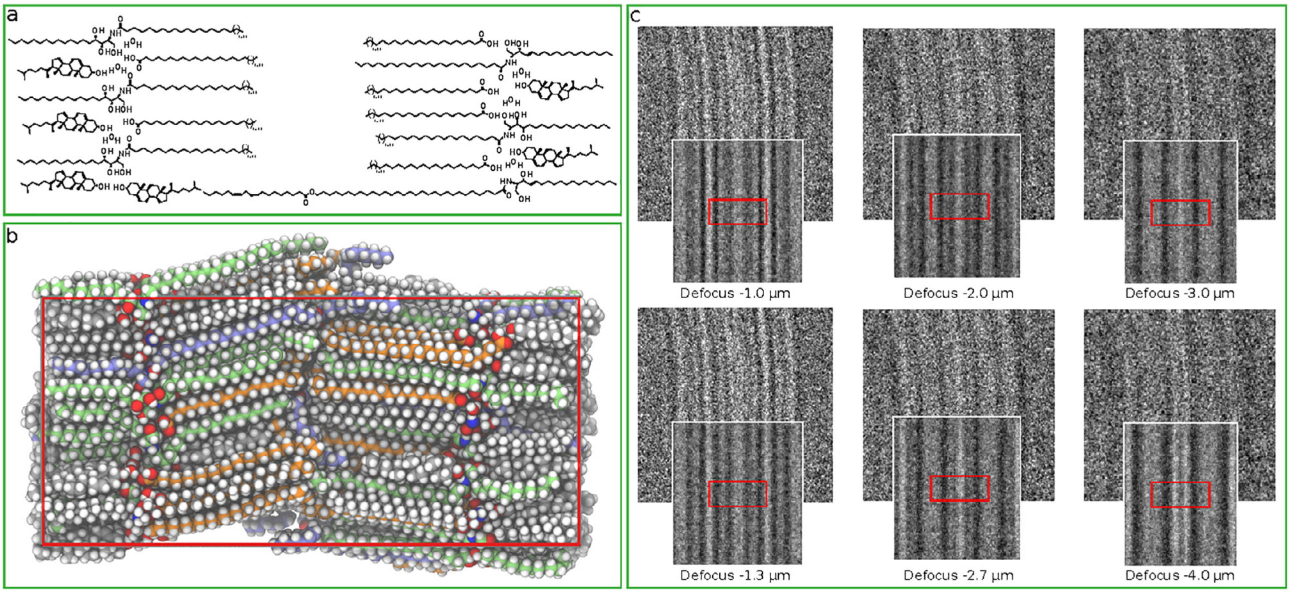

Fig. 4. The optimized MD model system for the stratum corneum lipid structure. a) shows a schematic drawing of the model system, where the CER fatty acid and FFA chain lengths are just included as a range. b) shows the model system after the MD simulation production phase. All carbon atoms in CER are green, in CER EOS light blue, in FFA orange and in CHOL grey. The rest of the atoms are colored according to atom types (oxygen: red, nitrogen: blue, hydrogen: white). The red box represents the periodic box, or unit cell, used in the MD simulations. Atoms outside the periodic box appear on the opposite side. c) shows a comparison between the simulated EM data derived from the model system (situated above, with the red square corresponding to the periodic box in b)) and the experimental cryo-EM reference data (situated beneath) obtained from near-native skin (adapted from Iwai et al. (2012)), at all three defocus levels of the image defocus series (−1, −2, −3 µm). In the upper set of images in c) the defocus levels of the simulated images are set according to the nominal defocus of the microscope, whereas in the lower set of images the defocus levels of the simulated images have been adjusted to optimize the match. The experimental images at defocus levels −1, −2 µm have been scaled down by 15 and 11%, respectively, to keep the same periodicity at all defocus levels. A video showing the proposed model can be found in the Supplementary material. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ratio in human skin. It might also be that the high amount of C30 FFA in the simulated structures compensates for the absence of FFA longer than C30, i.e. C32–C36, in the proposed model system, but which are present in stratum corneum (Norlén et al., 1998). The lower water permeability of the system without extra C30 FFA makes it interesting to further study that composition.

4.3. Calculated model water permeability

The here proposed model system (33/33/33/75/5/0.3), with an optimal fit between simulated and original EM data, had one of the lowest calculated water permeabilities of all the studied model systems, and its permeability could be even further reduced by removing the CER EOS from the model (Table 2). The proposed model's calculated water log K_P of $-4.5\ \mathrm{cm\,h^{-1}}$ was lower than the *in vitro/ex vivo* measured water log K_P of human skin in diffusion cells of $-3.0\ \mathrm{cm\,h^{-1}}$ to $-3.3\ \mathrm{cm\,h^{-1}}$ at 298 K (Scheuplein, 1965; Scheuplein and Blank, 1973), corresponding to an average log K_P of $-2.9\ \mathrm{cm\,h^{-1}}$ at 303 K (Abraham and Martins, 2004). The calculated permeability coefficients are based only on the intercellular lipid structure in the stratum corneum, which constitutes the main barrier. In diffusion cells the skin membrane is soaked in water, which will increase its permeability. It has been shown that a steady state is achieved only after approximately three days, at which point the permeability coefficient for pentanol can be increased by approximately 50% (Scheuplein, 1965) and for water and urea permeabilities have been shown to be affected even more (Van and Ackermann, 1987). It is expected that partial skin membrane hydration starts soon after the skin's exposure to water in the diffusion cell, making it difficult to compare *in vitro* with *in vivo* permeabilities (Van and Ackermann, 1987). Accordingly, the calculated water permeability of the model system with an increased number of water molecules per lipid (1.3 instead of 0.3) corresponds well with the reported diffusion cell experimental results (Scheuplein, 1965; Scheuplein and Blank, 1973), which is in agreement with the expected increased skin water permeability induced by water exposure in *in vitro* diffusion cell experiments. Other factors contributing to the higher measured water permeability of human skin *in vitro* compared with the here proposed MD model could be preparation induced microscopic defects in the skin samples *in vitro*, such as the presence of cracks or pores at former hair locations, or that the microstructure of the skin samples has been altered upon skin isolation by heating or excision and/or by storage in a freezer.

Adding acyl ceramide EOS increased the model system's water permeability. This effect was possibly due to a slightly increased lipid chain mobility and possibly also to the introduction of polar ester groups allowing favorable interactions with water. Speculatively, an increased water permeability may be the price to pay for a locally increased hydrocarbon chain mobility at the interface between the stacked lipid bilayers, allowing for an improved lateral displacement/sliding along the bilayer plane to accommodate bending of the tightly packed crystalline-like (gel) system.

4.4. Calculated model thermotropic behaviour

Differential scanning calorimetry (DSC) thermograms of isolated stratum corneum has shown four major transitions near 40 °C (T1), 75 °C (T2), 85 °C (T3) and 100 °C (T4) (Babita et al., 2006). However, the occurrence of the transition T1 (near 40 °C) has been reported to be variable and absent in stratum corneum lipid extracts, and no change in skin permeability has been observed at temperatures above T1 and below T2 (Cornwell and Barry, 1993). During simulated heating of our proposed model system (33/33/33/75/5/0.3), as well as of model systems with similar composition, no clear changes in lipid headgroup area or lamellar periodicity were observed near 40 °C (T1) (Fig. S23).

The second transition T2 (near 70 °C) and third transition T3 (near 85 °C) have, although extensively debated, either been attributed to

Table 2
Calculated permeability coefficients of different model systems to water. The permeability coefficients (log K_P in cm h^{-1}) are presented with one standard error given as the uncertainty. The Sign. diff. column indicates if the change is significant or not, at 95% confidence, compared to the 33/33/33/75/5/0.3 system (the system we propose as the most realistic one and which is marked in bold). The experimental log K_P of isolated human skin to water is approximately −2.9 at 303 K (−3.00 (Scheuplein, 1965) and −3.30 (Scheuplein and Blank, 1973) measured at 298 K). Model system IDs are based on their relative composition in molar % (../../../../../..): (total CER content/total CHOL content/total FFA content/relative amount of CHOL on CER sphingoid side/total CER EOS content/number of H_2O per lipid headgroup). The 33-hairpin/33/33 stacked bilayer no water model is a pure 1:1:1 CER:CHOL:FFA system with only CER NP C24 and FFA C24 chain lengths. The 33-hairpin/33/33 bilayer with water model is the same lipid system, but with water on both sides of the bilayer. The 33-np/33/33/75/5/0.3 model has no CER NS and the 33-ns/33/33/75/5/0.3 model has no CER NP. The 33/33-CHOL-SULF-10/33/80/5/0.3 is a system with 10% of the CHOL fraction replaced by cholesterol sulfate, the 33/33/33-charged-FFA-10/80/5/0.3 is a system with 10% of the FFA replaced by deprotonated FFA. The 33-C20-50/33/33/75/5/0.3 model contains 50% C20 and 50% C18 CER sphingoid chains. The 33/33/33-noXFFA30/75/5/0.3 model has no extra FFA of chain length C30.

System	log $K_{P_{calc}}$	Sign. diff.
Splayed bilayer model (Iwai et al., 2012)	−3.8 ± 0.2	Y
33-hairpin/33/33 stacked bilayer no water	−3.3 ± 0.1	Y
33-hairpin/33/33 bilayer with water	−1.4 ± 0.1	Y
33/33/33/100/0/0	−5.5 ± 0.1	Y
33/33/33/100/5/0.3	−4.6 ± 0.2	N
33/33/33/75/0/0.3	−5.3 ± 0.2	Y
33/33/33/75/5/0	−3.6 ± 0.3	Y
33/33/33/75/5/0.3	**−4.5 ± 0.2**	**N/A**
33/33/33/75/5/0.7	−4.8 ± 0.1	N
33/33/33/75/5/1.0	−3.7 ± 0.1	Y
33/33/33/75/5/1.3	−2.9 ± 0.2	Y
33/33/33/75/5/1.7	−4.2 ± 0.1	N
33/33/33/75/5/2	−4.0 ± 0.1	Y
33/33/33/75/5/3	−2.6 ± 0.1	Y
33/33/33/75/10/0.3	−3.5 ± 0.2	Y
33/33/33/70/5/0.3	−3.8 ± 0.2	Y
33/33/33/70/5/0.7	−4.9 ± 0.2	N
33-np/33/33/75/5/0.3	−4.2 ± 0.1	N
33-ns/33/33/75/5/0.3	−4.3 ± 0.1	N
33-C20-50/33/33/75/5/0.3	−4.5 ± 0.2	N
33/33/33/80/5/0.3	−4.9 ± 0.2	N
33/33-CHOL-SULF-10/33/80/5/0.3	−3.7 ± 0.1	Y
33/33/33-charged-FFA-10/80/5/0.3	−3.6 ± 0.3	Y
33/33/33-noXFFA30/75/5/0.3	−5.2 ± 0.1	Y

lipid gel to liquid crystalline phase transitions, or to orthorhombic to hexagonal lipid chain transitions, in different lipid domains, or lipid subpopulations, within isolated stratum corneum (Babita et al., 2006). In our model system slight changes in lipid headgroup area or lamellar periodicity were observed near 70 °C (T2) and near 85 °C (T3) (Fig. S23). It is, however, possible that the errors of the observables from the MD simulations are underestimated, e.g. due to long correlation times, and therefore that no changes in lamellar periodicity and headgroup area can be detected in the T2 and T3 temperature range. It is concluded that the calculated thermotropic behaviour of our model system is compatible with DSC data from isolated stratum corneum but do not offer a complete atomic-level physical explanation for the likely complex T2 and T3 endothermic transitions of skin.

The fourth endothermic transition T4 (near 100 °C) is thermally irreversible and has been assigned to denaturation of a stratum corneum protein component (Babita et al., 2006). During heating, a sharp and dramatic change in calculated lipid headgroup area and lamellar periodicity was observed in our proposed model system (33/33/33/75/5/0.3) near 100 °C (T4), corresponding to a melting of the model's gel-state lamellar lipid organization into a disordered state. All model systems with similar composition behaved similarly (Fig. S23). We conclude that the fourth endothermic transition T4 (near 100 °C) recorded in DSC experiments on isolated stratum corneum may not, as previously thought, correspond to protein denaturation, but to an irreversible melting of the intercellular lipid structure.

4.5. Comparison with RuO_4 skin staining patterns

Electron micrographs of plastic embedded RuO_4 stained skin show a characteristic broad-narrow-broad band staining pattern corresponding to the stratum corneum lipid structure (Madison et al., 1987). It is expected that the lipid structure's RuO_4-staining pattern mainly corresponds to the distribution of the most electronegative atoms in the system, i.e., the oxygens and nitrogens, although other chemical factors, such as local degree of RuO_4 reduction that may depend on local lipid chain unsaturation (Hill and Wertz, 2003), may also contribute. To obtain a rough estimate of what the electron microscopy intensity pattern of the optimized model system we compared the distribution of oxygens and nitrogens in the model system with the RuO_4-staining pattern of the stratum corneum's lipid structure in skin (Madison et al., 1987) (Fig. 5).

4.6. Artifacts in vitreous cryo-sections

It was observed in the reference cryo-electron micrograph defocus

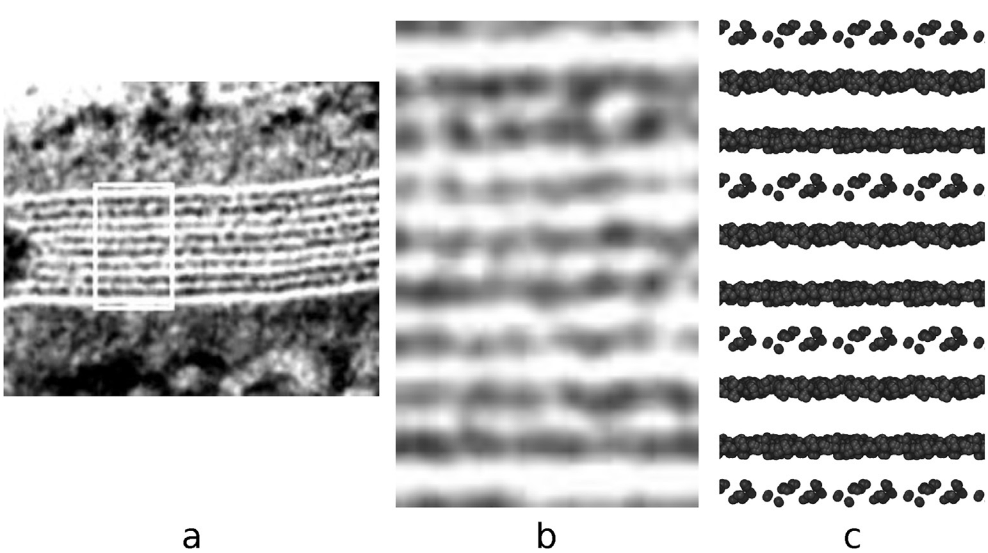

Fig. 5. Electron density pattern of the stratum corneum intercellular space in skin stained with ruthenium tetroxide (RuO_4). a) The RuO_4 staining pattern of the stratum corneum intercellular space of mouse skin (Madison et al., 1987). b) An enlarged view of the area marked by a white box in a). c) A van der Waals representation, scaled by a factor 2.5, of oxygens and nitrogens (the most electronegative atoms in the system) of the optimized lipid model system. This is a rough estimate of what the electron density pattern of the model system would look like after RuO_4 staining. The magnified image from the electron micrograph is scaled to match the van der Waals representation of the heavy atoms of the MD simulation model. Figures a) (and b)) were graciously obtained from Dr. Philip Wertz and are adapted from Madison et al. (1987), with permission.

series that the periodicity of the band pattern was gradually reduced at higher defocus levels (Iwai et al., 2012). The defocus series was acquired using repeated electron beam exposures at the same specimen location while stepwise increasing the microscope defocus, so the diminished periodicity was correlated with an increased accumulated electron dose to which the cryo-section had been exposed. Using defocus −1 μm as a reference, the shrinkage was 6% at defocus −2 μm and 15% at defocus −3 μm, or oppositely, using defocus −3 μm as reference, the elongation was 13% and 18%, respectively, at defocus levels −2 μm and −1 μm (herein compensated for by scaling those images down by 11% and 15%). It is well known that cryo-sectioning of vitrified tissue samples causes a compression of the obtained sections in the cutting direction (y) causing mainly a thickening, but also a widening, of the section in the z and x directions (Al-Amoudi et al., 2005; Han et al., 2008). In order to avoid artifactual compression of the lamellar repeat pattern of the stratum corneum's lipid structure during cryo-sectioning, the skin cryo-section from which the reference defocus series was obtained was cut along the lamellar plane of the lipid structure (Iwai et al., 2012). Widening of the cryo-section, to accommodate section compression along the cutting direction is likely and may have resulted in an artifactual slight increase of the lamellar periodicity of the lipid structure of the skin section. During electron beam exposure the compressed cryo-section may relax towards its original shape, resulting in an elongation in the cutting (length) direction and a shrinkage in the thickness and width dimensions. The shorter periodicity of the lipid structure observed in the last image (obtained at −3 μm defocus) of the defocus series may thus be closer to the original periodicity in the skin sample before widening imposed by cryo-sectioning. Dimensions of biostructures observed in cryo-electron micrographs of vitreous sections should therefore, although preserved in their near-native state, be regarded as approximate, which should be kept in mind when interpreting cellular cryo-EM data.

4.7. Determining precise defocus-levels in cryo-electron micrographs of vitreous sections

During data collection using cryo-EM of vitreous tissue sections, nominal defocus values may slightly differ from actual defocus values. In practice, actual defocus values are difficult to estimate exactly and may vary from nominal defocus values within a range of ±1 μm, despite good calibration of the microscope. Taking this into account, we simulated EM images for the optimized model at stepwise increased microscope defocus levels with a step-size of 0.1 μm, from 0.0 μm to −5.0 μm defocus. In this way correspondence between simulated and original cryo-EM data could be even further improved and, at the defocus levels of −1.3, −2.7, −4.0 μm, simulated and original data became close to indistinguishable for the optimized model system, as judged by visual inspection (Fig. 4c). We therefore speculate that the actual defocus values of the original reference defocus series were in fact −1.3, −2.7, −4.0 μm, rather than the given nominal −1, −2, −3 μm values (Iwai et al., 2012).

5. Conclusions

In the present paper we have used MD modelling combined with cryo-EM to analyse the molecular structure and function of the human skin's permeability barrier.

Starting from a compositionally simplified MD model based on the *splayed bilayer model* (Iwai et al., 2012), we have, by stepwise structural and compositional modifications, arrived at a thermodynamically stable MD model expressing simulated EM patterns matching original cryo-EM patterns from skin extremely closely. Strikingly, the closer the individual MD model's lipid composition was to that reported in human stratum corneum, the better was the match between the MD model's EM simulation patterns and the original cryo-EM patterns. Moreover, the closest-matching MD model's calculated water permeability and thermotropic behaviour were found to be compatible with that of human skin.

The new knowledge on the detailed structure and composition of the skin's permeability barrier, along with the availability of MD simulation, will facilitate comprehensive physics-based skin permeability calculations using more realistic models than have previously been available (Das et al., 2009; Gupta et al., 2016). This may aid predicting properties of drugs interacting with the skin and optimizing them for percutaneous drug delivery. In addition, it may be used for skin toxicity assessment. The effects and mechanisms of skin permeability enhancing formulations may also be investigated and optimized *in silico*.

From a methodological perspective, the here proposed procedure for molecular dynamics based analysis of cellular cryo-electron microscopy data might be applied to other biomolecular systems.

6. Author contributions

The molecular systems were generated by M.L. and A.N. The MD simulations were run by M.L. and C.W. Permeability calculations, and related method development, were performed by M.L. and C.W. The EM simulations were done by A.N. The project and experiments were planned by L.N., M.L., C.W., A.N., E.L. and B.D. The article was written by M.L., L.N., C.W., B.D., E.L. and A.N.

7. Declaration of conflicts of interest

This work is presented, in parts, in an application for a European patent with application number PCT/EP2017/076237 and title "SKIN PERMEABILITY PREDICTION".

The work presented herein was essentially funded by ERCO Pharma AB. L.N., M.L., A.N. and C.W. have stock options in ERCO Pharma AB. M.L., A.N. and C.W. are employed by ERCO Pharma AB.

Acknowledgements

The MD simulations were performed on resources provided by the Swedish National Infrastructure for Computing (SNIC) at PDC Centre for High Performance Computing. ERCO Pharma AB, the Welander foundation, Svenska läkaresällskapet, Oriflame, Shiseido Japan, LEO Pharma, the Wenner-Gren foundation and Vetenskapsrådet are acknowledged for generous financial support. We want to thank Berk Hess for valuable suggestions and ideas regarding permeability calculations.

Appendix A. Supplementary data

The supplementary material includes more detailed Methods and Results sections as well as a video illustrating the proposed model system (33/33/33/75/5/0.3) and ceramide force field parameters. Supplementary data associated with this article can be found, in the online version, athttp://dx.doi.org/10.1016/j.jsb.2018.04.005.

References

Abraham, M.H., Martins, F., 2004. Human skin permeation and partition: general linear free-energy relationship analyses. J. Pharm. Sci. 93, 1508–1523. http://dx.doi.org/10.1002/jps.20070.

Abraham, M.J., Murtola, T., Schulz, R., Páll, S., Smith, J.C., Hess, B., Lindahl, E., 2015. GROMACS: high performance molecular simulations through multi-level parallelism from laptops to supercomputers. SoftwareX 1–2, 19–25. http://dx.doi.org/10.1016/j.softx.2015.06.001.

Al-Amoudi, A., Chang, J.J., Leforestier, A., McDowall, A., Salamin, L.M., Norlén, L.P.O., Richter, K., Blanc, N.S., Studer, D., Dubochet, J., 2004. Cryo-electron microscopy of vitreous sections. EMBO J. 23, 3583–3588. http://dx.doi.org/10.1038/sj.emboj.7600366.

Al-Amoudi, A., Norlen, L.P.O., Dubochet, J., 2004. Cryo-electron microscopy of vitreous sections of native biological cells and tissues. J. Struct. Biol. 148, 131–135. http://dx.doi.org/10.1016/j.jsb.2004.03.010.

Al-Amoudi, A., Studer, D., Dubochet, J., 2005. Cutting artefacts and cutting process in vitreous sections for cryo-electron microscopy. J. Struct. Biol. 150, 109–121. http://

dx.doi.org/10.1016/j.jsb.2005.01.003.

Babita, K., Kumar, V., Rana, V., Jain, S., Tiwary, A.K., 2006. Thermotropic and spectroscopic behavior of skin: relationship with percutaneous permeation enhancement. Curr. Drug Deliv. 3, 95–113. http://dx.doi.org/10.2174/156720106775197466.

Björklund, S., Nowacka, A., Bouwstra, J.A., Sparr, E., Topgaard, D., 2013. Characterization of stratum corneum molecular dynamics by natural-abundance 13c solid-state NMR. PLoS One 8, e61889. http://dx.doi.org/10.1371/journal.pone.0061889.

Blank, I.H., Moloney, J., Emslie, A.G., Simon, I., Apt, C., 1984. The diffusion of water across the stratum corneum as a function of its water content. J. Invest. Dermatol. 82, 188–194. http://dx.doi.org/10.1111/1523-1747.ep12259835.

Bouwstra, J.A., Gooris, G.S., van der Spek, J.A., Bras, W., 1991. Structural investigations of human stratum corneum by small-angle X-ray scattering. J. Invest. Dermatol. 97, 1005–1012. http://dx.doi.org/10.1111/1523-1747.ep12492217.

Bouwstra, J.A., Gooris, G.S., Bras, W., Downing, D.T., 1995. Lipid organization in pig stratum corneum. J. Lipid Res. 36, 685–695.

Bouwstra, J.A., Gooris, G.S., Dubbelaar, F.E., Weerheim, A.M., Ijzerman, A.P., Ponec, M., 1998. Role of ceramide 1 in the molecular organization of the stratum corneum lipids. J. Lipid Res. 39, 186–196.

Bouwstra, J., Pilgram, G., Gooris, G., Koerten, H., Ponec, M., 2001. New aspects of the skin barrier organization. Skin Pharmacol. Appl. Skin Physiol. 14, 52–62. http://dx.doi.org/10.1159/000056391.

Bouwstra, J.A., Gooris, G.S., Dubbelaar, F.E., Ponec, M., 2002. Phase behavior of stratum corneum lipid mixtures based on human ceramides: the role of natural and synthetic ceramide 1. J. Invest. Dermatol. 118, 606–617. http://dx.doi.org/10.1046/j.1523-1747.2002.01706.x.

Bouwstra, J.A., de Graaff, A., Gooris, G.S., Nijsse, J., Wiechers, J.W., van Aelst, A.C., 2003. Water distribution and related morphology in human stratum corneum at different hydration levels. J. Invest. Dermatol. 120, 750–758. http://dx.doi.org/10.1046/j.1523-1747.2003.12128.x.

Breathnach, A.S., Goodman, T., Stolinski, C., Gross, M., 1973. Freeze-fracture replication of cells of stratum corneum of human epidermis. J. Anat. 114, 65–81.

Chen, L.Y., 2008. Nonequilibrium fluctuation-dissipation theorem of Brownian dynamics. J. Chem. Phys. 129, 144113. http://dx.doi.org/10.1063/1.2992153.

Cornwell, P., Barry, B., 1993. The routes of penetration of ions and 5-fluorouracil across human skin and the mechanisms of action of terpene skin penetration enhancers. Int. J. Pharm. 94, 189–194. http://dx.doi.org/10.1016/0378-5173(93)90023-9.

Crooks, G.E., 1999. Entropy production fluctuation theorem and the nonequilibrium work relation for free energy differences. Phys. Rev. E 60, 2721–2726. http://dx.doi.org/10.1103/PhysRevE.60.2721.

Crooks, G.E., 2000. Path-ensemble averages in systems driven far from equilibrium. Phys. Rev. E 61, 2361–2366. http://dx.doi.org/10.1103/PhysRevE.61.2361.

Dahlén, B., Pascher, I., 1979. Molecular arrangements in sphingolipids. Thermotropic phase behaviour of tetracosanoylphytosphingosine. Chem. Phys. Lipids 24, 119–133. http://dx.doi.org/10.1016/0009-3084(79)90082-3.

Damien, F., Boncheva, M., 2010. The extent of orthorhombic lipid phases in the stratum corneum determines the barrier efficiency of human skin in vivo. J. Invest. Dermatol. 130, 611–614. http://dx.doi.org/10.1038/jid.2009.272.

Das, C., Olmsted, P.D., Noro, M.G., 2009. Water permeation through stratum corneum lipid bilayers from atomistic simulations. Soft Matter 5, 4549–4555. http://dx.doi.org/10.1039/B911257J.

de Sousa Neto, D., Gooris, G., Bouwstra, J., 2011. Effect of the ω-acylceramides on the lipid organization of stratum corneum model membranes evaluated by X-ray diffraction and FTIR studies (Part I). Chem. Phys. Lipids 164, 184–195. http://dx.doi.org/10.1016/j.chemphyslip.2010.12.007.

Eichner, A., Sonnenberger, S., Dobner, B., Hau, T., Schroeter, A., Neubert, R.H., 2016. Localization of methyl-branched ceramide [EOS] species within the long-periodicity phase in stratum corneum lipid model membranes: a neutron diffraction study. Biochim. Biophys. Acta 1858, 2911–2922. http://dx.doi.org/10.1016/j.bbamem.2016.09.002.

Eichner, A., Stahlberg, S., Sonnenberger, S., Lange, S., Dobner, B., Ostermann, A., Schrader, T.E., Hau, T., Schroeter, A., Huster, D., Neubert, R.H., 2017. Influence of the penetration enhancer isopropyl myristate on stratum corneum lipid model membranes revealed by neutron diffraction and 2H NMR experiments. Biochim. Biophys. Acta 1859, 745–755. http://dx.doi.org/10.1016/j.bbamem.2017.01.029.

Elias, P.M., Friend, D.S., 1975. The permeability barrier in mammalian epidermis. J. Cell Biol. 65, 180–191. http://dx.doi.org/10.1083/jcb.65.1.180.

Garson, J.C., Doucet, J., Lévêque, J.L., Tsoucaris, G., 1991. Oriented structure in human stratum corneum revealed by X-ray diffraction. J. Invest. Dermatol. 96, 43–49. http://dx.doi.org/10.1111/1523-1747.ep12514716.

Gómez-Fernández, J.C., Villalaín, J., 1998. The use of FT-IR for quantitative studies of the apparent pKa of lipid carboxyl groups and the dehydration degree of the phosphate group of phospholipids. Chem. Phys. Lipids 96, 41–52. http://dx.doi.org/10.1016/S0009-3084(98)00079-6.

Grice, K., Sattar, H., Baker, H., 1972. The effect of ambient humidity on transepidermal water loss. J. Invest. Dermatol. 58, 343–346. http://dx.doi.org/10.1111/1523-1747.ep12540526.

Gupta, R., Sridhar, D.B., Rai, B., 2016. Molecular dynamics simulation study of permeation of molecules through skin lipid bilayer. J. Phys. Chem. B 120, 8987–8996. http://dx.doi.org/10.1021/acs.jpcb.6b05451.

Han, H.M., Zuber, B., Dubochet, J., 2008. Compression and crevasses in vitreous sections under different cutting conditions. J. Microsc. 230, 167–171. http://dx.doi.org/10.1111/j.1365-2818.2008.01972.x.

Hatta, I., Ohta, N., Inoue, K., Yagi, N., 2006. Coexistence of two domains in intercellular lipid matrix of stratum corneum. Biochim. Biophys. Acta 1758, 1830–1836. http://dx.doi.org/10.1016/j.bbamem.2006.08.014.

Hill, J.R., Wertz, P.W., 2003. Molecular models of the intercellular lipid lamellae from epidermal stratum corneum. Biochim. Biophys. Acta 1616, 121–126. http://dx.doi.org/10.1016/S0005-2736(03)00238-4.

Hou, S.Y.E., Mitra, A.K., White, S.H., Menon, G.K., Ghadially, R., Elias, P.M., 1991. Membrane structures in normal and essential fatty acid-deficient stratum corneum: characterization by ruthenium tetroxide staining and X-ray diffraction. J. Invest. Dermatol. 96, 215–223. http://dx.doi.org/10.1111/1523-1747.ep12461361.

Humphrey, W., Dalke, A., Schulten, K., 1996. VMD: visual molecular dynamics. J. Mol. Graph. 14, 33–38. http://dx.doi.org/10.1016/0263-7855(96)00018-5.

Iwai, I., Han, H., Hollander, L.d., Svensson, S., Öfverstedt, L.G., Anwar, J., Brewer, J., Bloksgaard, M., Laloeuf, A., Nosek, D., Masich, S., Bagatolli, L.A., Skoglund, U., Norlén, L., 2012. The human skin barrier is organized as stacked bilayers of fully extended ceramides with cholesterol molecules associated with the ceramide sphingoid moiety. J. Invest. Dermatol. 132, 2215–2225.. http://dx.doi.org/10.1038/jid.2012.43.. URL: < https://creativecommons.org/licenses/by-nc-nd/3.0/legalcode > .

Jorgensen, W.L., Chandrasekhar, J., Madura, J.D., Impey, R.W., Klein, M.L., 1983. Comparison of simple potential functions for simulating liquid water. J. Chem. Phys. 79, 926–935. http://dx.doi.org/10.1063/1.445869.

Klauda, J.B., Venable, R.M., Freites, J.A., O'Connor, J.W., Tobias, D.J., Mondragon-Ramirez, C., Vorobyov, I., MacKerell, A.D., Pastor, R.W., 2010. Update of the CHARMM all-atom additive force field for lipids: validation on six lipid types. J. Phys. Chem. B 114, 7830–7843. http://dx.doi.org/10.1021/jp101759q.

Kosztin, I., Barz, B., Janosi, L., 2006. Calculating potentials of mean force and diffusion coefficients from nonequilibrium processes without Jarzynski's equality. J. Chem. Phys. 124, 064106. http://dx.doi.org/10.1063/1.2166379.

Kühlbrandt, W., 2014. Microscopy: cryo-EM enters a new era. eLife 3, 3. http://dx.doi.org/10.7554/eLife.03678. e03678.

Lieckfeldt, R., Villalaín, J., Gómez-Fernández, J., Lee, G., 1995. Apparent pKa of the fatty acids within ordered mixtures of model human stratum corneum lipids. Pharm. Res. 12, 1614–1617. http://dx.doi.org/10.1023/A:1016280714593.

Lundborg, M., Lindahl, E., 2014. Automatic GROMACS topology generation and comparisons of force fields for solvation free energy calculations. J. Phys. Chem. B 119, 810–823. http://dx.doi.org/10.1021/jp505332p.

Madison, K.C., Swartzendruber, D.C., Wertz, P.W., Downing, D.T., 1987. Presence of intact intercellular lipid lamellae in the upper layers of the stratum corneum. J. Invest. Dermatol. 88, 714–718. http://dx.doi.org/10.5555/uri:pii:0022202X87902648.

Masukawa, Y., Narita, H., Sato, H., Naoe, A., Kondo, N., Sugai, Y., Oba, T., Homma, R., Ishikawa, J., Takagi, Y., Kitahara, T., 2009. Comprehensive quantification of ceramide species in human stratum corneum. J. Lipid Res. 50, 1708–1719. http://dx.doi.org/10.1194/jlr.D800055-JLR200.

McIntosh, T.J., 2003. Organization of skin stratum corneum extracellular lamellae: diffraction evidence for asymmetric distribution of cholesterol. Biophys. J. 85, 1675–1681. http://dx.doi.org/10.1016/S0006-3495(03)74597-4.

Moghadam, S.H., Saliaj, E., Wettig, S.D., Dong, C., Ivanova, M.V., Huzil, J.T., Foldvari, M., 2013. Effect of chemical permeation enhancers on stratum corneum barrier lipid organizational structure and interferon alpha permeability. Mol. Pharm. 10, 2248–2260. http://dx.doi.org/10.1021/mp300441c.

Mojumdar, E., Groen, D., Gooris, G., Barlow, D., Lawrence, M., Deme, B., Bouwstra, J., 2013. Localization of cholesterol and fatty acid in a model lipid membrane: a neutron diffraction approach. Biophys. J. 105, 911–918. http://dx.doi.org/10.1016/j.bpj.2013.07.003.

Mojumdar, E.H., Gooris, G.S., Barlow, D.J., Lawrence, M.J., Deme, B., Bouwstra, J.A., 2015. Skin lipids: localization of ceramide and fatty acid in the unit cell of the long periodicity phase. Biophys. J. 108, 2670–2679. http://dx.doi.org/10.1016/j.bpj.2015.04.030.

Mojumdar, E.H., Gooris, G.S., Groen, D., Barlow, D.J., Lawrence, M.J., Demé, B., Bouwstra, J.A., 2016. Stratum corneum lipid matrix: location of acyl ceramide and cholesterol in the unit cell of the long periodicity phase. Biochim. Biophys. Acta 1858, 1926–1934. http://dx.doi.org/10.1016/j.bbamem.2016.05.006.

Nakazawa, H., Ohta, N., Hatta, I., 2012. A possible regulation mechanism of water content in human stratum corneum via intercellular lipid matrix. Chem. Phys. Lipids 165, 238–243. http://dx.doi.org/10.1016/j.chemphyslip.2012.01.002.

Norlén, L., Nicander, I., Lundsjö, A., Cronholm, T., Forslind, B., 1998. A new HPLC-based method for the quantitative analysis of inner stratum corneum lipids with special reference to the free fatty acid fraction. Arch. Dermatol. Res. 290, 508–516. http://dx.doi.org/10.1007/s004030050344.

Ohta, N., Ban, S., Tanaka, H., Nakata, S., Hatta, I., 2003. Swelling of intercellular lipid lamellar structure with short repeat distance in hairless mouse stratum corneum as studied by X-ray diffraction. Chem. Phys. Lipids 123, 1–8. http://dx.doi.org/10.1016/S0009-3084(02)00126-3.

Pham, Q.D., Björklund, S., Engblom, J., Topgaard, D., Sparr, E., 2016. Chemical penetration enhancers in stratum corneum – relation between molecular effects and barrier function. J. Control. Release 232, 175–187. http://dx.doi.org/10.1016/j.jconrel.2016.04.030.

Pilgram, G.S.K., Marjolein Engelsma-van Pelt, A., Koerten, H.K., Bouwstra, J.A., 1999. Electron diffraction provides new information on human stratum corneum lipid organization studied in relation to depth and temperature. J. Invest. Dermatol. 113, 403–409. http://dx.doi.org/10.1046/j.1523-1747.1999.00706.x.

Potts, R.O., Guzek, D.B., Harris, R.R., McKie, J.E., 1985. A noninvasive, in vivo technique to quantitatively measure water concentration of the stratum corneum using attenuated total-reflectance infrared spectroscopy. Arch. Dermatol. Res. 277, 489–495. http://dx.doi.org/10.1007/BF00510068.

Pouya, I., Pronk, S., Lundborg, M., Lindahl, E., 2017. Copernicus, a hybrid dataflow and peer-to-peer scientific computing platform for efficient large-scale ensemble sampling. Future Gener. Comput. Syst. 71, 18–31. http://dx.doi.org/10.1016/j.future.

2016.11.004.
Pronk, S., Páll, S., Schulz, R., Larsson, P., Bjelkmar, P., Apostolov, R., Shirts, M.R., Smith, J.C., Kasson, P.M., van der Spoel, D., Hess, B., Lindahl, E., 2013. GROMACS 4.5: a high-throughput and highly parallel open source molecular simulation toolkit. Bioinformatics 29, 845–854. http://dx.doi.org/10.1093/bioinformatics/btt055.
Pronk, S., Pouya, I., Lundborg, M., Rotskoff, G., Wesén, B., Kasson, P.M., Lindahl, E., 2015. Molecular simulation workflows as parallel algorithms: the execution engine of copernicus, a distributed high-performance computing platform. J. Chem. Theory Comput. 11, 2600–2608. http://dx.doi.org/10.1021/acs.jctc.5b00234.
Rullgård, H., Öfverstedt, L.G., Masich, S., Daneholt, B., Öktem, O., 2011. Simulation of transmission electron microscope images of biological specimens. J. Microsc. 243, 234–256. http://dx.doi.org/10.1111/j.1365-2818.2011.03497.x.
Scheuplein, R.J., 1965. Mechanism of percutaneous adsorption. J. Invest. Dermatol. 45, 334–346. http://dx.doi.org/10.1038/jid.1965.140.
Scheuplein, R.J., Blank, I.H., 1973. Mechanism of percutaneous absorption. Iv. Penetration of nonelectrolytes (alcohols) from aqueous solutions and from pure liquids. J. Invest. Dermatol. 60, 286–296. http://dx.doi.org/10.1111/1523-1747.ep12723090.
Schmitt, T., Lange, S., Dobner, B., Sonnenberger, S., Hauss, T., Neubert, R.H.H., 2017. Investigation of a CER and [AP] based stratum corneum modelling membrane system: using specifically deuterated CER together with a neutron diffraction approach. doi: 10.1021/acs.langmuir.7b01848.
Schröter, A., Kessner, D., Kiselev, M.A., Hau, T., Dante, S., Neubert, R.H.H., 2009. Basic nanostructure of stratum corneum lipid matrices based on ceramides [EOS] and [AP]: a neutron diffraction study. Biophys. J. 97, 1104–1114. http://dx.doi.org/10.1016/j.bpj.2009.05.041.
Schullery, S.E., Seder, T.A., Weinstein, D.A., Bryant, D.A., 1981. Differential thermal analysis of dipalmitoylphosphatidylcholine–fatty acid mixtures. Biochemistry 20, 6818–6824.
Školová, B., Hudská, K., Pullmannová, P., Kováčik, A., Palát, K., Roh, J., Fleddermann, J., Estrela-Lopis, I., Vávrová, K., 2014. Different phase behavior and packing of ceramides with long (C16) and very long (C24) acyls in model membranes: infrared spectroscopy using deuterated lipids. J. Phys. Chem. B 118, 10460–10470. http://dx.doi.org/10.1021/jp506407r.
Sparr, E., Wennerström, H., 2001. Responding phospholipid membranes-interplay between hydration and permeability. Biophys. J. 81, 1014–1028. http://dx.doi.org/10.1016/S0006-3495(01)75759-1.
Stahlberg, S., Lange, S., Dobner, B., Huster, D., 2016. Probing the role of ceramide headgroup polarity in short-chain model skin barrier lipid mixtures by 2H solid-state NMR spectroscopy. Langmuir 32, 2023–2031. http://dx.doi.org/10.1021/acs.langmuir.5b04173.
Stahlberg, S., Eichner, A., Sonnenberger, S., Kováčik, A., Lange, S., Schmitt, T., Demé, B., Hau, T., Dobner, B., Neubert, R.H.H., Huster, D., 2017. Influence of a novel dimeric ceramide molecule on the nanostructure and thermotropic phase behavior of a stratum corneum model mixture. Langmuir 33, 9211–9221. http://dx.doi.org/10.1021/acs.langmuir.7b01227.
Stone, J., 1998. An efficient library for parallel ray tracing and animation (Master's thesis). University of Missouri-Rolla.
Swartzendruber, D.C., Wertz, P.W., Kitko, D.J., Madison, K.C., Downing, D.T., 1989. Molecular models of the intercellular lipid lamellae in mammalian stratum corneum. J. Invest. Dermatol. 92, 251–257. http://dx.doi.org/10.1111/1523-1747.ep12276794.
Van Der Merwe, E., Ackermann, C., 1987. Physical changes in hydrated skin. Int. J. Cosmet. Sci. 9, 237–247. http://dx.doi.org/10.1111/j.1467-2494.1987.tb00478.x.
Venable, R., Sodt, A., Rogaski, B., Rui, H., Hatcher, E., MacKerell Jr., A., Pastor, R., Klauda, J., 2014. CHARMM all-atom additive force field for sphingomyelin: elucidation of hydrogen bonding and of positive curvature. Biophys. J. 107, 134–145. http://dx.doi.org/10.1016/j.bpj.2014.05.034.
Weerheim, A., Ponec, M., 2001. Determination of stratum corneum lipid profile by tape stripping in combination with high-performance thin-layer chromatography. Arch. Dermatol. Res. 293, 191–199. http://dx.doi.org/10.1007/s004030100212.
White, S.H., Mirejovsky, D., King, G.I., 1988. Structure of lamellar lipid domains and corneocyte envelopes of murine stratum corneum. An X-ray diffraction study. Biochemistry 27, 3725–3732. http://dx.doi.org/10.1021/bi00410a031.